

# **Certus User Guide**

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# About This Manual

The *Certus Closure Solution* software provides a design closure solution to perform distributed ECO on larger and complex designs. This manual describes how to install, configure, and use Certus to implement digital integrated circuits.

## Audience

This manual is written for experienced designers of digital integrated circuits. Such designers must be familiar with design planning, placement, and routing, block implementation, chip assembly, and design verification. Designers must also have a solid understanding of UNIX and Tcl/Tk programming.

## How This Manual Is Organized

The chapters in this manual are organized to follow the flow of tasks through the design process. Because of variations in design implementations and methodologies, the order of the chapters will not correspond to any specific design flow.

Each chapter focuses on the concepts and tasks related to the particular design phase or topic being discussed.

In addition, the following sections provide prerequisite information for using the Certus software:

- Installation and Startup
- Introduction to Certus
- Flows
- Manual ECO

# Conventions Used in This Manual

This section describes the typographic and syntax conventions used in this manual.

|             |                                                                                                                                                                                                                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>text</i> | <p>Indicates text that you must type exactly as shown. For example:</p> <pre>set_message -severity info</pre>                                                                                                                                                                                                            |
| <i>text</i> | <p>Indicates information for which you must substitute a name or value.</p> <p>In the following example, you must substitute the name of a specific file for <i>file_name</i>:</p> <pre>compare_release -out_file file_name</pre>                                                                                        |
| <i>text</i> | <p>Indicates the following:</p> <ul style="list-style-type: none"><li>• Text found in the graphical user interface (GUI), including form names, button labels, and field names</li><li>• Terms that are new to the manual, are the subject of discussion, or need special emphasis</li><li>• Titles of manuals</li></ul> |
| [ ]         | <p>Indicates optional arguments.</p> <p>In the following example, you can specify none, one, or both of the bracketed arguments:</p> <pre>command [-arg1] [arg2 value]</pre>                                                                                                                                             |
| [   ]       | <p>Indicates an optional choice from a mutually exclusive list.</p> <p>In the following example, you can specify any of the arguments or none of the arguments, but you cannot specify more than one:</p> <pre>command [arg1   arg2   arg3   arg4]</pre>                                                                 |
| {   }       | <p>Indicates a required choice from a mutually exclusive list.</p> <p>In the following example, you must specify one, and only one, of the arguments:</p> <pre>command {arg1   arg2   arg3}</pre>                                                                                                                        |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>{ [ ] [ ] }</code>                           | <p>Indicates a required choice of one or more items in a list.</p> <p>In the following example, you must choose one argument from the list, but you can choose more than one:</p> <pre>command {[arg1] [arg2] [arg3]}</pre>                                                                                                                                                                             |
| <code>{ }</code>                                   | <p>Indicates curly braces that must be entered with the command syntax.</p> <p>In the following example, you must type the curly braces:</p> <pre>command arg1 {x y}</pre>                                                                                                                                                                                                                              |
| <code>...</code>                                   | <p>Indicates that you can repeat the previous argument.</p>                                                                                                                                                                                                                                                                                                                                             |
| <code>.</code><br><code>.</code><br><code>.</code> | <p>Indicates an omission in an example of computer output or input.</p>                                                                                                                                                                                                                                                                                                                                 |
| <i>Command - Subcommand</i>                        | <p>Indicates a command sequence, which shows the order in which you choose commands and subcommands from the GUI menu.</p> <p>In the following example, you choose <i>Power</i> from the menu, then <i>Power Planning</i> from the submenu, and then <i>Add Ring</i> from the displayed list:</p> <p><i>Power - Power Planning - Add Ring</i></p> <p>This sequence opens the <i>Add Rings</i> form.</p> |

## Related Documents

For more information about Certus, see the following documents. You can access these and other Cadence documents using Doc Assistant (cloud-based) documentation system.

### Certus Product Documentation

- [What's New in Certus](#)  
Provides information about new features in this release of Certus.
- [Certus Text Command Reference](#)  
Describes the Certus text commands, including syntax and examples.



For a complete list of documents provided with this release, see the Doc Assistant online documentation system.

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# Installation and Startup

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- [Certus Product and Installation Information](#)
- [Setting Up the Certus Runtime Environment](#)
- [Launching the Certus Console](#)
- [Starting the Certus Software](#)
- [Accessing Certus Documentation and Help](#)

## Certus Product and Installation Information

For product, release, and installation information, see the README file at any of the following locations:

- <http://downloads.cadence.com>
- In the software installation, where it is also available when you are using or running the software

For information about the Certus licenses, see the [Certus Licensing Requirements](#).

### Related Topics

- [Installation and Startup](#)
- [Setting Up the Certus Runtime Environment](#)
- [Launching the Certus Console](#)
- [Starting the Certus Software](#)
- [Accessing Certus Documentation and Help](#)

# Setting Up the Certus Runtime Environment

If `install_dir` is the location of the Certus installation, then you should set up the runtime environment as follows:

- Add the `install_dir/bin` directory to the path. The bin directory has links to all the public executable files in the install hierarchy.
- If you want the Certus man pages to be available with the Unix man command, then you can add the `install_dir/share/certus/man` to the MANPATH envvar.
- If you want the Stylus Common UI Certus man pages to be available with the Unix man command, then you can add `install_dir/share/certus/stylus` to the MANPATH envvar.
- If you want the Tcl man pages to be available with the Unix man command, then you can add the `install_dir/share/tcltools/man` to the MANPATH envvar.

For example, you can add the following to the startup shell script:

```
set install_dir = /tools/certus22.1/lrx86 set path = ($install_dir/bin $path)
setenv MANPATH $install_dir/share/certus/man:$install_dir/share/tcltools/ man:$MANPATH
```

## Note:

When Certus launches, it automatically adds the legacy or Stylus CUI man pages (if the `-stylus` parameter is specified), and the Tcl man pages at the beginning of the current MANPATH inside Certus. Therefore, within Certus, the `man` command will show both the sets of man pages before any other man pages.

## Temporary File Locations

Each Certus session creates a temporary directory to store temporary files at the beginning of the run.

By default, `ssv_tmpdir` is created in the `/tmp` directory. If the Unix envvar `TMPDIR` is set, `ssv_tmpdir` is created inside `$TMPDIR`.

The name of the `tmp_dir` will appear as given below:

```
ssv_tmpdir_\[pid\]_xxxxxx
```

Where `_xxxxxx` is a string that is added to make the directory unique. For example:

```
ssv_tmpdir_10233_nfp9ez
```

The temporary directory is automatically removed on exit or if the run is terminated with a catchable signal (for example, SIGSEGV).

## Related Topics

- [Installation and Startup](#)
- [Certus Product and Installation Information](#)
- [Launching the Certus Console](#)
- [Starting the Certus Software](#)
- [Accessing Certus Documentation and Help](#)

# Launching the Certus Console

When you start Certus in the non-GUI environment using the `-no_gui` parameter, the UNIX window (shell, xterm, and so on) where you start the Certus session is called the Certus console. This is where you enter all Certus text commands and where the software displays messages. When a session is active, this console displays the following prompt:

```
certus>
```

When you start Certus in the GUI environment, the Certus console is displayed within the main Certus window. If you use this console for other actions, for example, to use the vi editor, the session suspends until you finish the action.

If you suspend the session by typing Control-z, the `certus>` prompt is no longer displayed.

To return to the Certus session, type `fg`, which brings the session to the foreground.

## Related Topics

- [Installation and Startup](#)
- [Certus Product and Installation Information](#)
- [Setting Up the Certus Runtime Environment](#)
- [Starting the Certus Software](#)
- [Accessing Certus Documentation and Help](#)

## Starting the Certus Software

You can start the Certus software using the executable: `certus`

Note that the binary pointed to by Certus is still the SSV binary structure. For example,

```
/install_dir/bin/certus
```

Using the above command, Certus will start Tempus clients using Tempus and Certus clients using Innovus.

**Note:** By default, Certus runs in ECO mode.

### Related Topics

- [Installation and Startup](#)
- [Certus Product and Installation Information](#)
- [Setting Up the Certus Runtime Environment](#)
- [Launching the Certus Console](#)
- [Accessing Certus Documentation and Help](#)

## Accessing Certus Documentation and Help

You can access the Certus documentation and help system by using the following methods:

- Launching Doc Assistant
- Accessing Documentation and Help from Certus GUI
- Using the Certus man and help Commands on the Command-Line
- Other Sources of Cadence Product Information

### Launching Doc Assistant

To launch the Doc Assistant tool, you can use one of the following ways:

- Linux systems: Navigate to the `<certus_install_directory>/lnx86/bin` directory, and run the `cda` command.

- Tool's Help menu.

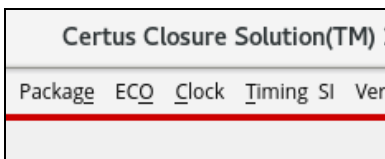
## Accessing Documentation and Help from Certus GUI

You can access documentation from the Certus GUI using the *Help* menu.

### Select Help on the Main Certus Menu

Follow the steps given below:

#### Figure 1-1 Help Menu



→ Select *Help*, and click any of the following options:

- *Text Command Reference*

Opens the Table of Contents page of the text command reference.

- *User Guide*

Opens the Table of Contents page of the user guide.

- *What's New*

Opens the Table of Contents page of the What's new document.

## Using the Certus man and help Commands on the Command-Line

### Using the *help* Command to View the Command Syntax

- To see syntax information for a command, type the following command in the software console:

```
help command_name
```

For example, to see syntax information for the `set_eco_opt_mode` command, type the following command:

```
help set_eco_opt_mode
```

The software displays the following text:

Usage: `set_eco_opt_mode`

- To see the entire list of Certus commands and their syntax, type the following command in the

software console:

```
help
```

### ***Using the man Command to View the Command Description***

To see the complete set of information for a Certus Legacy command, type the following command in the software console:

```
man command_name
```

For example, to see the complete set of information for the `set_eco_opt_mode` command, type the following command:

```
man set_eco_opt_mode
```

### ***Using the help Command to View Message Summary***

To see the message summary of a particular message ID, type the following command in the software console:

```
help msg_id
```

For example, to see the message summary for the `TECHLIB-1002` message ID, type the following command:

```
help TECHLIB-1002
```

### ***Using the man Command to View Message Detail***

To see the message details of a particular message ID, type the following command in the software console:

```
man msg_id
```

For example, to see the message summary for the `SI-2253` message ID, type the following command:

```
man SI-2253
```

The detailed description is not available for all active message IDs.

## **Other Sources of Cadence Product Information**

You can also get help on Cadence products by selecting *Customer Support* on the *Help* menu. The *Customer Support* submenu provides access to the following Cadence resources:

- **Cadence Online Support**  
Opens Cadence Online Support in your browser.
- **Web Collaboration**  
Opens your default browser and explains the current methodology Cadence follows to support web-based collaboration.

## Related Topics

- [Installation and Startup](#)
- [Certus Product and Installation Information](#)
- [Setting Up the Certus Runtime Environment](#)
- [Launching the Certus Console](#)
- [Starting the Certus Software](#)



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# Introduction to Certus

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- [Certus Product Overview](#)
- [Certus Licensing Requirements](#)
- [Certus Inputs Requirements](#)
- [Certus Template Scripts](#)
- [Recommended Guidelines for Certus](#)

## Certus Product Overview

The *Certus Closure Solution* software, also known as Certus, is a next-generation design closure solution that enables engineers to perform distributed ECO on larger and complex designs. It provides enhanced performance and capacity by using the Distributed MMMC architecture, which fully leverages multiple threads across distributed processes, enabling maximum scalability. Certus provides a flexible computing environment that enables a single user to perform subsystem/full-chip ECO timing/power closure, reducing turnaround time from several days to hours.

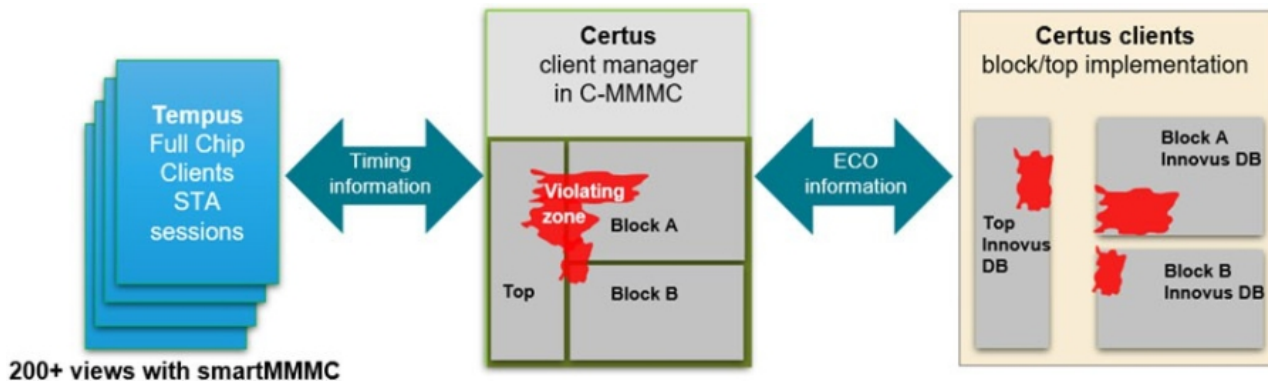
Another advantage of this solution is that Certus allows the users to perform ECO routing and parasitic extraction in batch mode, to collect updated parasitic, and perform updated timing analysis with signoff accuracy post ECO within the same session.

Currently, users run timing analysis and ECO closure in separate sessions when the design has a large number of MMMC views. After the ECO closure run, users need to implement the ECOs, collect new Verilog and SPEF data, and then perform post-ECO timing analysis.

The Certus software addresses these usability concerns by automating all those steps through a single cockpit and a single session.

Other advanced ECO solutions, such as High Capacity ECO (which reduces the database size) and SmartMMMC (which compacts the timing graph when a large number of timing views are provided), are complementary to Certus.

The figure below illustrates the architecture where Certus is the main cockpit and uses Certus clients to implement ECOs at the top and block levels, and Tempus clients to collect signoff timing.



## Certus Solutions Architecture Overview

Certus includes the following key features:

- Ensures that each partition and top level loads its own database.
- Reduced memory peak as physical data is no longer assembled in the Tempus client manager session.
- Reduced ECO cycle time, thus enabling improved productivity. This is due to the lighter design-loading phase and the embedded ECO implementation, which allows iteration within the same session.
- Includes a simple use model where physical data are loaded directly from Innovus DB. There is no requirement to provide a full hierarchical UPF (power intent comes from STA clients and Innovus DBs).
- Drives implementation directly from the Certus cockpit, where ECO implementation is done in batch mode on Certus clients. In addition, updated parasitic data is redirected to the Tempus STA session for final timing analysis post ECO implementation.
- Provides support for multi-technology LEF design architecture.
- Allows ECO optimization for inter-DIE paths concurrently, where each DIE is from different process nodes. This is specific to 3DIC designs.
- Compatible with advanced existing features of Tempus, such as Distributed MMMC ECO, smartMMMC, and High Capacity ECO.

### Related Topics

- [Certus Licensing Requirements](#)
- [Certus Inputs Requirements](#)

- [Certus Template Scripts](#)
- [Recommended Guidelines for Certus](#)

## Certus Licensing Requirements

The Certus software follows the sequence below for checking out licenses:

1. To invoke Certus, the software checks out the client manager license - Certus Optimization Client Manager (CRTS200).
2. To launch Tempus clients within Certus, the Tempus license per client is checked out.
3. To launch Certus clients, the software checks out one (CRTS150) license per Certus client:
  - a. For a 3nm technology node, each Certus client checks out a Certus 3nm license (CRTS230).
  - b. For a 2nm technology node, each Certus client checks out a Certus 2nm license (CRTS232).
4. To run ECO routing, the software checks out the Innovus Base (INVS) per Certus client and Innovus Technology (INVS\_tech) licenses.
5. To run signoff parasitic extraction, the software checks out the Quantus licenses.

You can iterate back to Step 1; however, all the licenses (except Quantus and Pegasus) will remain checked out until the Certus client manager exits.

### Related Topics

- [Certus Product Overview](#)
- [Certus Inputs Requirements](#)
- [Certus Template Scripts](#)
- [Recommended Guidelines for Certus](#)

## Certus Inputs Requirements

The input data for setting up the distributed ECO environment in Certus is as follows:

- Tempus STA saved sessions:

- These can be C-MMMC or SMMC - non-physical sessions.
- Identify which view should be used for power reporting.
- Innovus DB pointer for top-only and each partition:
  - Ensure that Innovus DBs are based on the same netlist as STA sessions.
  - You can use ILM inside Innovus DB, while the same modules are flattened inside STA sessions.
- Resources:
  - 1 client manager machine (16 CPUs or more preferred).
  - 1 machine per STA sessions (16 CPUs or more preferred).
  - Extra machines - when the number of Innovus DBs is much higher than the number of STA sessions.

## Related Topics

- [Certus Product Overview](#)
- [Certus Licensing Requirements](#)
- [Certus Template Scripts](#)
- [Recommended Guidelines for Certus](#)

# Certus Template Scripts

Use the template scripts below to run Certus in different modes:

## Script 1 - Certus Distributed Hierarchical Timing Closure - Full Flat Mode

The script below performs distributed hold optimization in full-flat mode and ends with the ECO implementation step. This step performs ECO routing and parasitic extraction for each Certus client, followed by a timing update with the updated parasitics on the timing clients.

```
set_multi_cpu_usage -localCpu 24 -remoteHost 8 -cpuPerRemoteHost 4
set_distribute_host <your LSF args> -timeOut 10000
set eco_setup_view_list {setup_1v setup_2v setup_3v setup_4v}
set eco_hold_view_list {hold_1v hold_2v hold_3v hold_4v}
set eco_dynamic_view {setup_3v} ; set eco_leakage_view {setup_3v}
```

## File Pointing to all Innovus DBs Opened for ECO

```
set_eco_opt_mode -distributed_opt_block_restore_file blocks.txt
```

## Loads the STA Sessions on Tempus Clients

```
distribute_read_design -restore_design all -restore_dir \ TIMING_SESSIONS/ -outdir  
reports
```

## Sets the Distributed Options for Certus Clients

```
set_multi_cpu_usage -remoteHost <equal_to_partition_count> \  
-cpuPerRemoteHost <value>  
set_distribute_host <your LSF args> -timeOut 10000
```

## Starts ECO Clients

```
start_distributed_eco_clients
```

## Runs Hold Fixing in PBA Mode

```
eco_opt_design -hold
```

## Performs ecoRoute/extraction and Parasitic Back-annotation

```
implement_distributed_eco
```

## Collect Final Signoff Timing from Tempus Clients and Save the Final Innovus DB

```
time_design_signoff  
export_distributed_eco -out_dir Final -export_only innovus_dbs
```

## Script 2- Certus Distributed Hierarchical Timing Closure - Inter-partition High Capacity ECO Mode

The script below performs Hold timing closure by enabling existing High capacity ECO for large sub-systems and top level designs. It optimizes the logical inter-partition, meaning that each timing path crosses the boundary of one of the listed partitions. The script ends with the ECO implementation step, which performs ECO routing and parasitic extraction on each Certus client, followed by a timing update with updated parasitics on the timing clients.

```
set_multi_cpu_usage -localCpu 24 -remoteHost 8 -cpuPerRemoteHost 4
set_distribute_host <your LSF args> -timeOut 10000
set_eco_setup_view_list {setup_1v setup_2v setup_3v setup_4v}
set_eco_hold_view_list {hold_1v hold_2v hold_3v hold_4v}
set_eco_dynamic_view {setup_3v} ; set_eco_leakage_view {setup_3v}
```

### File Pointing to all Innovus DBs Opened for ECO

```
set_eco_opt_mode -distributed_opt_block_restore_file blocks.txt
```

### Reduces the DB around partition-to-partition and between top-and-partition

```
set_distribute_mmmc_mode -enable_high_capacity_eco
set_eco_opt_mode -inter_partition {blockA blockB}
```

### Loads the STA Sessions on Tempus Clients

```
distribute_read_design -restore_design all -restore_dir \ TIMING_SESSIONS/ -outdir
reports
```

### Sets the Distributed Options for Certus Clients

```
set_multi_cpu_usage -remoteHost <equal_to_partition_count> \
-cpuPerRemoteHost <value>
set_distribute_host <your LSF args> -timeOut 10000
```

### Starts the Certus Clients

```
start_distributed_eco_clients
```

## Runs Hold Fixing in PBA Mode

```
eco_opt_design -hold
```

## Performs ecoRoute/extraction and Parasitic Back-annotations

```
implement_distributed_eco
```

## Collects Final Signoff Timing from Tempus Clients and Saves the Final Innovus DB

```
time_design_signoff  
export_distributed_eco -out_dir Final -export_only innovus_dbs
```

## Script 3- Distributed Hierarchical Power Reduction - With ECO Implementation and Parasitic Back-annotation Loop

The script below performs Power recovery by enabling existing High capacity ECO for large subsystems and top-level designs. It optimizes the logical inter-partition, meaning that each timing path crosses the boundary of exactly one of the listed partitions. The script does the ECO implementation step, which performs ECO routing and parasitic extraction on each Certus client, followed by a timing update with updated parasitics on the timing clients. Next, the Setup timing recovery using Vth swapping only is performed to address any remaining timing violations that would have resulted from extensive ECO during power optimization.

```
set_multi_cpu_usage -localCpu 24 -remoteHost 8 -cpuPerRemoteHost 4  
set_distribute_host <your LSF args> -timeOut 10000  
set eco_setup_view_list {setup_1v setup_2v setup_3v setup_4v}  
set eco_hold_view_list {hold_1v hold_2v hold_3v hold_4v}  
set eco_dynamic_view {setup_3v} ; set eco_leakage_view {setup_3v}  
set_eco_opt_mode -distributed_opt_block_restore_file blocks.txt  
set_distribute_mmmc_mode -enable_high_capacity_eco  
set_eco_opt_mode -inter_partition {blockA blockB}  
distribute_read_design -restore_design all -restore_dir TEMPUS_SESSIONS/ -outdir  
reports  
set_multi_cpu_usage -remoteHost <equal_to_partition_count> \  
-cpuPerRemoteHost <value>  
set_distribute_host <your LSF args> -timeOut 10000  
start_distributed_eco_clients  
set_eco_opt_mode -max_slack 1000  
set_eco_opt_mode -allow_multiple_incremental true
```

```
set_eco_opt_mode -verbose false
eco_opt_design -power
implement_distributed_eco
```

## Performs Setup Timing Recovery to Fix any Timing Gaps Post ECO Implementation

```
set_eco_opt_mode -setup_recovery true -max_slack 0
eco_opt_design -setup
```

### Related Topics

- [Certus Product Overview](#)
- [Certus Licensing Requirements](#)
- [Certus Inputs Requirements](#)
- [Recommended Guidelines for Certus](#)

## Recommended Guidelines for Certus

Few important guidelines that you need to follow while using the Certus tool. These are mentioned below:

- Reserve temporary space for Tempus and Certus clients, approximately one-third of the tool's memory capacity, as mentioned in the examples below:

- If the STA session uses 600GB of memory, reserve 200GB of temporary memory:

```
set_distribute_host -lsf -resource "mem>600000 rusage\  
[mem=600000\:tmp=200000\] span\[hosts=1\]" -args { -q xxx -W 100:00 -P tempus  
-n 16} -timeOut 90000
```

- If Innovus clients use 80GB memory, reserve 20GB temporary memory:

```
set_distribute_host -lsf -resource "mem>80000 rusage\[mem=80000\:tmp=20000\  
span\[hosts=1\]" -args { -q xxx -W 100:00 -P innovus -n 8} -timeOut 90000
```

If you intend to reuse Tempus client machines for Certus clients, ensure you increase the memory request for the Tempus clients.

- Ensure that your `$PATH` environment variable points to the same Innovus version as the Certus client manager session, such as Innovus 26.1x and Certus 26.1x.
- Ensure that the STA saved sessions in the `viewDefinition` file have the Quantus technology file defined in `create_rc_corner`.



- Ensure that the STA session flat netlist matches the Innovus DB netlist.
  - Any mismatch will indicate don't touch and can impact timing or power closure.
  - Ensure that the netlist is not flattened in STA sessions and Innovus DB.
- To perform a full flat power reporting on the power view, use the settings below:  

```
distribute_command -views <powerViewName> {report_power}
```
- Use the same tool to generate initial SPEF as Certus clients will re-extract parasitic:
  - In case you are generating SPEF on an Innovus DB on a design with ILM, ensure that you add `unflattenIlm` before doing `extractRC`.
- Ensure that your Quantus command file does not contain a hard-coded output file name (since the same file name will be executed on each Certus client), and the input file name is accessible from where the Innovus DB is loaded.
  - Provide UPF/CPF in STA sessions when the design has multiple supply voltages.
  - When running power optimization on inter-partitions, the power numbers printed by Certus are for the logic covering the inter-partition zone only and not the full design power.
  - Ensure that the top-only level is always opened for ECO (which is a part of the block restore file) - in case it is not a case of abutted partitions with no logic at the top.
  - The `start_distributed_eco_clients` command does not prevent the rest of the script from being executed.
    - This command is to start the Innovus distributed clients, but allows the next command to be processed.
    - The purpose is to have `eco_opt_design` start generating ECO DB while all Innovus clients are booted in parallel.
    - Do not perform any netlist change between `start_distributed_eco_clients` and `eco_opt_design`.
  - In case of an error, you should first check the client manager/client machines, verify the temporary disk space, ensure that there is no CPU overloading situation, and check the current disk space.
    - Check the `host-monitor.log` file to determine the health of the manager machine and each of the Tempus client machines created for your run.

- Check the `host-monitor-eco-clients.log` file to determine the health of the manager machine and each of the Certus client machines created for your run.
- Ensure to apply the correct views for the Setup analysis and Hold analysis.
  - Adding Hold only views to the Setup active view list may lead to unexpected Setup timing.
- When having Innovus DBs generated by `saveTestcase`, those cannot be loaded by distributed ECO:
  - Recommended solution is to initially use `saveDesign -libs`, which replaces all functions of `saveTestcase`, or you need to load each of those Innovus DBs individually, and save them again with `saveDesign`.
- You cannot use CUI STA with LUI Innovus DB (and vice-versa):
  - You can generate a CUI STA or CUI Innovus DB by loading a LUI DB in CUI and saving it.
- It is recommended to load each Innovus DB in a standalone Innovus session to check its validity before starting Certus:
  - Run `checkPlace-noPreplaced`: Verify that there are no placement violations.
  - Run `ecoRoute`: Verify that the tool can converge quickly and does not end with too many DRCs.
- In case of any error or SEGV that the client manager is reporting from the clients, you must review the client's log files:
  - Tempus Clients: Checks in the directory pointed by `distribute_read_design`.
  - Certus Clients: Checks in the directory pointed by `set_eco_opt_mode - distributed_opt_out_dir xxx`.
- In case you observe significant timing jumps in `time_design_signoff` after `implement_distributed_eco`, do the following:
  - Check the Certus client log files to verify that there is no large instance movement during placement legalization.
  - Ensure that the same parasitic extraction tools or options that were utilized to create the original parasitic data used for the initial STA are enabled.
  - Perform a dry run using the below-mentioned flow to verify that timing does not

change during the implementation phase without any ECOs being performed:

- Load design
  - `time_design_signoff`
  - `implement_distributed_eco -force`
  - `time_design_signoff`
- To enable custom resource distribution related to CPU and memory configurations for Certus and Tempus clients, use the below settings when running the `set_eco_opt_mode - custom_distribution_file <file>` command parameter.
  - For every client, either for Tempus or Certus, or Tempus and Certus, specify the following:

```
module=<moduleName> view=<viewName> nCpu=<Number of threads allowed> {<farm distribution instruction>}
```
  - For client machine sharing, specify the View as `view=<viewName>` with one (or multiple) Modules, `module=<moduleName>`
  - For two Certus clients sharing the same machine, specify `module=<moduleName>` twice on the same line (where the `nCPU` value is then divided by the number of Certus clients and provided to each Certus client).
  - **Note:**
    - When the `-custom_distribution_file <file>` parameter is specified, Certus ignores the `-resource` and `-args` parameters of `set_distribute_host` from the main TCL file and uses only those settings that are defined within `<file>`. The `-timeOut` parameter of the `set_distribute_host` command will be honored to specify the timeout for a submitted job that will never start. After a timeout, a warning message is displayed.
    - The tool generates an error if any required view or module is not listed in it. This means the user should list all views/modules in this file.
    - The tool first displays the specific error (which can vary in each case, depending on what the user provided incorrectly). However, an informational message is displayed as shown below:

```
Usage: The format for each line: module=<blockName> view=<viewName> nCpu=<INT> {bsub ...}. Each line should contain a module, or a view, or
```

both. One or more modules can be specified in a line, but only one view can be specified. The nCpu and {bsub ...} are mandatory and can appear only once in a line. The host launching string should be enclosed within '{' and '}' and should be at the trailing end of a line.

- This approach allows users the flexibility to assign different clients to different queues.
- The timeout period is specified using the `set_distribute_host -timeOut` command parameter in the main TCL script.
- In case of CMMMC Tempus clients, the client would take the last occurrence of any of the active views being specified.

### Examples:

- The following command refers to a file (`distribution.tcl`) where each client requests a separate machine:

```
set_eco_opt_mode -custom_distribution_file distribution.tcl
```

where (distribution.tcl) file includes:

```
module=topChip nCpu=16 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o
/dev/null -n 8 -R 'select\[OSREL==EE80\] rusage\[mem=250000,tmp=50000\]
span\[hosts=1\]'}
module=small_cpu nCpu=8 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o
/dev/null -n 8 -R 'select\[OSREL==EE80\] rusage\[mem=150000,tmp=50000\]
span\[hosts=1\]'}
module=big_GPU nCpu=16 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o
/dev/null -n 16 -R 'select\[OSREL==EE80\] rusage\[mem=250000,tmp=50000\]
span\[hosts=1\]'}
view=setup_view1 nCpu=16 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o
/dev/null -n 16 -R 'select\[OSREL==EE80\] rusage\
[mem=200000,tmp=100000\] span\[hosts=1\]'}
view=setup_view2 nCpu=16 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o
/dev/null -n 16 -R 'select\[OSREL==EE80\] rusage\
[mem=200000,tmp=100000\] span\[hosts=1\]'}
view=setup_hold1 nCpu=8 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o
/dev/null -n 8 -R 'select\[OSREL==EE80\] rusage\[mem=200000,tmp=100000\]
span\[hosts=1\]'}
view=setup_power nCpu=8 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o
/dev/null -n 8 -R 'select\[OSREL==EE80\] rusage\[mem=200000,tmp=100000\]
span\[hosts=1\]'}

```

- The following command refers to a file (`distribution.tcl`) that clients use when sharing machines.

`set_eco_opt_mode -custom_distribution_file distribution.tcl`

where (`distribution.tcl`) file includes:

```
module=topChip view=setup_view1 nCpu=16 {bsub -q large -W 100:00 -P
SSV:26.1:Certus -o /dev/null -n 8 -R 'select\[OSREL==EE80\] rusage\
[mem=450000,tmp=150000\] span\[hosts=1\]'}
module=small_cpu view=setup_hold1 nCpu=8 {bsub -q large -W 100:00 -P
SSV:26.1:Certus -o /dev/null -n 8 -R 'select\[OSREL==EE80\] rusage\
[mem=350000,tmp=150000\] span\[hosts=1\]'}
module=big_GPU view=setup_view2 nCpu=16 {bsub -q large -W 100:00 -P
SSV:26.1:Certus -o /dev/null -n 16 -R 'select\[OSREL==EE80\] rusage\
[mem=450000,tmp=150000\] span\[hosts=1\]'}
module=mini_CPU1 module=mini_CPU2 view=setup_view3 nCpu=16 {bsub -q
large -W 100:00 -P SSV:26.1:Certus -o /dev/null -n 16 -R 'select\
[OSREL==EE80\] rusage\[mem=450000,tmp=150000\] span\[hosts=1\]'}
view=setup_power nCpu=8 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o
/dev/null -n 8 -R 'select\[OSREL==EE80\] rusage\[mem=200000,tmp=100000\]
span\[hosts=1\]'}
module=mega_GPU nCpu=32 {bsub -q large -W 100:00 -P SSV:26.1:Certus -o
/dev/null -n 32 -R 'select\[OSREL==EE80\] rusage\
[mem=350000,tmp=150000\] span\[hosts=1\]'}

```

## Related Topics

- [Certus Product Overview](#)
- [Certus Licensing Requirements](#)
- [Certus Inputs Requirements](#)
- [Certus Template Scripts](#)

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# Flows

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- [Hierarchical Top+Boundary Model ECO Flow](#)
- [Template Scripts](#)

## Hierarchical Top+Boundary Model ECO Flow

The *Hierarchical Top+Boundary Model ECO* flow in Certus can be used to perform timing and power closure across interface paths while loading lower-level models/blocks and closing a higher-level module/block. This flow analyzes the design hierarchically using boundary constraints at the top level, thus allowing faster and more efficient timing analysis. It also reduces memory usage and runtime significantly on larger designs.

When the *High Capacity ECO (HC-ECO)* flow with Top+BM is used, the `timing_enable_boundary_model_eco` command is applied before the boundary model is generated and loaded. This is done to ensure that the boundary model can be used for ECO.

Based on the user-defined settings, the Top+BM database is optimized using any of the following methods:

- Reduction based on the timing violations seen during the Top+BM STA session (Setup/Hold/DRV).
- Reduction based on the list of partitions to be optimized for inter\_partition timing paths.
- Reduction based on the user-defined path groups and a list of DRV-violated nets or pins.

To use this flow, Certus follows the approach below:

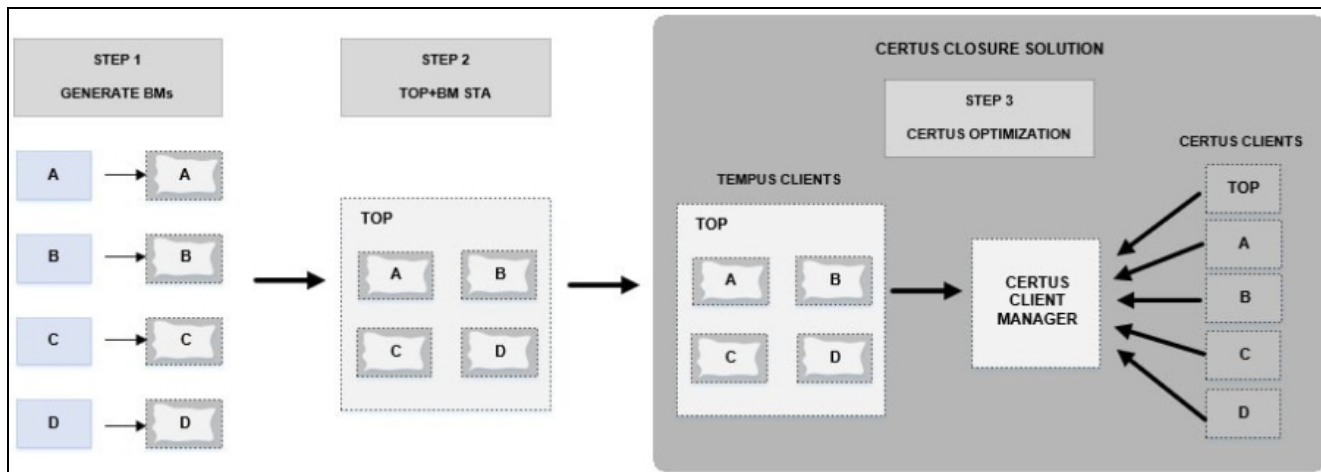
- *Top+BM HC-ECO Using Certus Environment*

### Top+BM HC-ECO Using Certus Environment

Using this approach, the tool:

- Performs DRV+Setup+Hold timing closure on violating zone.
- Performs Top+BM timing closure using High Capacity ECO inside Certus.

The following figure describes the hierarchical Top+BM ECO flow steps:



### Top+BM HC-ECO (Using Certus Environment)

- **Step 1:** Run Tempus at block level to generate boundary models.
- **Step 2:** Run Tempus to perform STA on Top+BM.
- **Step 3:** Certus optimizes TOP and the boundary of each partition.

### Related Topics

- [Template Scripts](#)

## Template Scripts

- [Template 1](#)
- [Template 2](#)
- [Template 3](#)

### Template 1

The Certus flow performs hold timing closure on a Top+BM database based on timing violations from the Top+BM STA clients. You can decide to enable either GBA or PBA mode.

### BM Generation Session

The BM session can be SMSC, MMMC, or D-MMMC.

```
set timing_full_context_flow 1
set timing_sum_cc_for_all_small_aggressors 1
set timing_write_aggressors_to_disk 1
set timing_enable_model_accurate_nef 1
set timing_enable_filter_fanin_cone 1
timing_enable_boundary_model_eco
...
create_boundary_model -overwrite -no_extended_fanin \
-latch_effort_low -skip_constants -exclude_no_delay_port \
-dir BM_${BlockName}_${view} -overwrite
The STA saved session can be SMSC or MMMC.
```

## Top+BM STA Session

```
timing_enable_boundary_model_eco
<load_Top+BM_design>
save_design view1.enc
```

## Certus Session

```
setDistributeHost -local -timeout 10000
set_multi_cpu_usage -localCpu 24 -remoteHost 4 -cpuPerRemoteHost 4
set_distribute_mmmc_mode -enable_high_capacity_eco
set eco_setup_view_list {setup_1v setup_2v setup_3v setup_4v}
set eco_hold_view_list {hold_1v hold_2v hold_3v hold_4v}
set eco_dynamic_view {setup_3v} ; set eco_leakage_view {setup_3v}
set_eco_opt_mode -distributed_opt_block_restore_file blocks.txt
set_eco_opt_mode -optimize_replicated_modules true
distribute_read_design -restore_design all -restore_dir \
STA_211/ -outdir reports_Paradime
set_distribute_host <your LSF args> -timeout 10000
start_distributed_eco_clients
eco_opt_design -hold
```

## Template 2

The Certus flow performs inter-partition Hold timing closure on a Top+BM database using a user-defined partition. The boundary model generation flow will remain the same as specified in Template 1.

The STA saved session can be SMSC or MMMC. The same template can be used to perform the inter-partition Power recovery.



## Top+BM STA Session

```
timing_enable_boundary_model_eco  
<load_Top+BM_design>  
save_design view1.enc
```

## Certus Session

```
setDistributeHost -local -timeout 10000  
set_multi_cpu_usage -localCpu 24 -remoteHost 4 -cpuPerRemoteHost 4  
set_distribute_mmmc_mode -enable_high_capacity_eco  
set_eco_opt_mode -distributed_opt_block_restore_file blocks.txt  
set eco_setup_view_list {setup_1v setup_2v setup_3v setup_4v}  
set eco_hold_view_list {hold_1v hold_2v hold_3v hold_4v}  
set eco_dynamic_view {setup_3v} ; set eco_leakage_view {setup_3v}  
set_eco_opt_mode -inter_partition {blockA blockB}  
set_eco_opt_mode -intra_partition {}  
set_eco_opt_mode -optimize_replicated_modules true  
distribute_read_design -restore_design all -restore_dir \  
STA_211/ -outdir reports_Paradime  
set_distribute_host <your LSF args> -timeout 10000  
start_distributed_eco_clients  
eco_opt_design -hold
```

## Template 3

The Certus flow performs DRV, Setup, and Hold timing closure on a Top+BM database using the user-defined Setup/Hold path\_groups and a list of DRV pins. The boundary model generation flow will remain the same as specified in Template 1.

The STA saved session can be SMSC or MMMC.

## Top+BM STA Session

```
timing_enable_boundary_model_eco  
<load_Top+BM_design>  
save_design view1.enc
```

## Certus Session

```
setDistributeHost -local -timeout 10000  
set_multi_cpu_usage -localCpu 24 -remoteHost 4 -cpuPerRemoteHost 4  
set_distribute_mmmc_mode -enable_high_capacity_eco  
set_eco_opt_mode -distributed_opt_block_restore_file blocks.txt  
set_eco_opt_mode -path_group_file_name {TCL script which creates mySetupPG and myHoldPG  
path_groups}
```

```
set_eco_opt_mode -include_setup_path_groups {mySetupPG}
set_eco_opt_mode -include_hold_path_groups {myHoldPG}
set_eco_opt_mode -select_drv_pin_file {myDRVPinList.txt}
set eco_setup_view_list {setup_1v setup_2v setup_3v setup_4v}
set eco_hold_view_list {hold_1v hold_2v hold_3v hold_4v}
set eco_dynamic_view {setup_3v} ; set eco_leakage_view {setup_3v}
set_eco_opt_mode -optimize_replicated_modules true
distribute_read_design -restore_design all -restore_dir \
STA_211/ -outdir reports_Paradime
set_distribute_host <your LSF args> -timeout 10000
start_distributed_eco_clients
eco_opt_design -drv; eco_opt_design -setup; eco_opt_design -hold
```

## Related Topics

- [Certus Licensing Requirements](#)
- [Certus Inputs Requirements](#)
- [Certus Template Scripts](#)
- [Hierarchical Top+Boundary Model ECO Flow](#)
- [Recommended Guidelines for Certus](#)

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# Manual ECO

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- [Manual ECO Overview](#)

## Manual ECO Overview

Manual ECO in Certus fixes timing, functional, or design rule violations in a design after signoff stages. This is especially useful in critical design phases, enabling precise fixes and reducing turnaround time before final signoff.

There are two modes supported in Manual ECO:

- Non-Interactive mode
- Interactive mode

### Non-Interactive Mode

In this mode, the tool sources the specified Innovus-compliant ECO files. The command specified in these ECO files will first be executed on the respective Certus clients before being processed for execution on the manager. After this, the processed commands will be executed (block by block) on both the manager and the Tempus clients.

Note that if an error occurs at any stage during execution of the ECO files, processing will stop and control will be returned to the Certus manager.

To enable non-interactive mode, use the `replay_distributed_eco - block_eco_file_list <file>` setting, which specifies the list of Certus client block module names and the corresponding ECO files as input. You can also specify multiple ECO files for the same module.

## Sample Template Script:

```
set_multi_cpu_usage -localCpu 12 -remoteHost 2 -cpuPerRemoteHost 8
## Settings for Certus start up
set_eco_opt_mode -distributed_opt_block_restore_file
/icd/etsproject3/qor/RAK/LUI/blkRestore.txt
set_distribute_mmmmc_mode -enable_high_capacity_eco
distribute_read_design -restore_design $dbnames -outdir paradime_Output
set_multi_cpu_usage -localCpu 12 -remoteHost 3 -cpuPerRemoteHost 8
start_distributed_eco_clients
#Setting to replay Certus Manual ECO
replay_distributed_eco -block_eco_file_list <path to block_eco_file> ./block_eco_file
```

Below is the template for `block_eco_file_list`:

```
dtmf_recvr_core ECO/drv_dtmf_recvr_core_eco_innovus.tcl
ptn_wrapper ECO/drv_ptn_wrapper_eco_innovus.tcl
tdsp_core ECO/drv_tdsp_core_eco_innovus.tcl
dtmf_recvr_core ECO/setup_dtmf_recvr_core_eco_innovus.tcl
ptn_wrapper ECO/setup_ptn_wrapper_eco_innovus.tcl
tdsp_core ECO/setup_tdsp_core_eco_innovus.tcl
```

## Interactive Mode

In this mode, you need to provide the Tempus-compliant ECO commands on the Certus Client Manager (CCM), which then modifies them to obtain the relevant block-level ECOs (based on the instance on which ECO is performed). The processed ECO command is further sent to the Certus client through the TCL command interface.

The Certus client runs this ECO command and generates an enhanced ECO command for CCM (for example, by appending corner spec data), which CCM then processes. Further, CCM executes this ECO command on all the Tempus Clients first, followed by the Certus Client Manager.

### Note:

- The interactive manual ECO does not include support for eval.
- You should not specify more than 1k ECOs to be sourced manually.

## Sample Template Script:

```
set_distribute_mmmc_mode -eco true -merge_view_definitions
set_multi_cpu_usage -localCpu 12 -remoteHost 2 -cpuPerRemoteHost 8
## Settings for Certus start up
set_eco_opt_mode -distributed_opt_block_restore_file
/icd/etsproject3/qor/RAK/LUI/blkRestore.txt
distribute_read_design -restore_design {save_db_dmmmc_setup_func
save_db_dmmmc_setup_test} -outdir paradime_Output
set_multi_cpu_usage -localCpu 12 -remoteHost 3 -cpuPerRemoteHost 8
start_distributed_eco_clients
##Tempus Compliant Manual ECO commands
change_cell -inst TEST_CONTROL_INST/g143 -cell NAND2X2
add_repeater -cell CLKBUF2 -name FE_ESOC2_n_206 -term { CPF_ISO_HIER_INST_107/g471/A }
delete_repeater -inst PM_INST/FE_ESOC76_n_41
```

For details related to Manual ECO commands, refer to the *Manual ECO Commands* chapter of the *Tempus Text Command Reference Manual*.

## Related Topics

- [Certus Product Overview](#)
- [Certus Template Scripts](#)